🧠 STAGE 3 — ANTI-SCRAPING AWARENESS (FULL REVISION NOTES)

🎯 SYSTEM GOAL (LOCK THIS FIRST)

- Jab scraper break ho:
  - **panic nahi**
  - **code turant change nahi**
  - **pehle problem ka naam lagana**
- Goal = 👉 *"Why it broke?" samajhna before "How to fix?"*

1️⃣ Detection Awareness — *How sites catch bots*

🔑 Core Idea

- Website **tumhara Python code nahi dekhti**
- Website sirf **HTTP requests ka behaviour** dekhti hai

🧠 Website kya judge karti hai

- Request **speed**
- Request **timing**
- **Regularity / exact pattern**
- **Repetition**
- **Consistency**

👉 Agar behaviour **human jaisa nahi**, to bot suspect.

👀 Real-life analogy

- Human = ruk-ruk ke kaam karta hai
- Bot = machine gun jaisa exact speed

🚨 Important rules

- Code clean hona ≠ safe
- Selectors perfect hona ≠ safe
- **Behaviour > Code**

🧩 Problem this solves

- 403 / empty HTML / random failures ka real reason samajh aata hai
- Galat jagah debugging band hoti hai

2️⃣ Browser vs Script Traffic

🔑 Core Idea

- Website ye dekhti hai:
  - **Browser jaisa traffic** hai ya
  - **Script jaisa naked traffic**

🧑 Browser traffic (Human)

- Multiple requests jaati hain:
  - HTML
  - CSS
  - JS
  - Images
  - API calls
- Cookies set hoti hain
- Navigation hota hai
- History hoti hai

👉 Website bole: *"Normal user"*

🤖 Script traffic (Bot)

- Ek direct `GET`
- No cookies
- No history
- No context

- No assets

👉 Website bole: *"Suspicious"*

🚫 Important clarification

- Browser ko **copy karna goal nahi**
- Browser ko **study karna goal hai**

🧠 Key rule

```
Browser = conversation
Script = demand
```

3️⃣ Request Patterns Matter More Than Code

🔑 Core Idea

- Website **average speed** nahi
- Website **bursts + rhythm** dekhti hai

🔍 Request pattern kya hota hai

- Order (kaun pehle)
- Timing (gap kitna)
- Frequency (kitni baar)
- Consistency (kitna robotic)
- Context (session grow ho rahi?)

⚠️ Dangerous misunderstanding

- "1 sec sleep hai → safe hoon" ❌
- Burst andar ho to **still unsafe**

🧠 Key rule

```
200 OK + empty HTML = pattern issue
```

🐞 Async ka truth

- Async = waiting time hide karta hai
- Server ko **bursts dikhti hain**
- Blind async = ban magnet

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4 Soft Block vs Hard Block

🟤 Hard Block

- Clear rejection
- Examples:
  - 403 Forbidden
  - 429 Too Many Requests
  - Redirect to error / captcha page

👉 Website chillati hai: *"NO"*

🟡 Soft Block (MOST DANGEROUS)

- Status code = 200 OK
- HTML bahut chhota
- Data missing
- Selectors return `None`

👉 Website smile karti hai aur **data chhupa leti hai**

🧠 Golden Rule

```
403 / 429 → Hard block
200 + empty HTML → Soft block
```

⚠️ Why soft block dangerous

- Code crash nahi hota

- Pipeline quietly corrupt hoti hai
- Tum galat data save kar sakte ho

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5 Rate-Based Bans

🔑 Core Idea

- "Too many requests" = **per unit time**
- Total count irrelevant hai

🚨 Rate-based ban triggers

- Bursts (multiple requests ek saath)
- High concurrency repeat
- Exact timing (clock-aligned)

🧠 Async + rate-ban

- Async khud problem nahi
- **Blind async problem hai**

🧩 Real rule

```
Smooth flow > Burst flow
```

🐞 Common misunderstanding

- "Sleep hai phir bhi ban kyu?" → sleep batch ke beech hai, request ke beech nahi

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6 Header Fingerprinting

🔑 Core Idea

- Website **headers ka combination** dekhti hai
- Ye combination = **fingerprint**

🚫 Beginner mistake

- Browser se headers copy-paste ❌

⚠️ Suspicious cases

- Bahut kam headers
- Bahut perfect headers
- Har request me same headers
- Headers aur behaviour ka mismatch

🧠 Important rule

```
Missing headers = suspicious
Perfect headers = suspicious
```

🔍 Browser reality

- Header values mostly stable hoti hain
- Context change hota hai:
  - cookies
  - referer
  - request order
  - navigation

7️⃣ Cookies & Sessions

🔑 Core Idea

- Cookies = memory
- Session = relationship over time

❌ Stateless requests

- Har request new
- No cookies

- No continuity

👉 Website bole: *“Har baar naya banda?”*

☑ Session-based requests

- Cookies reused
- Same identity
- Natural flow

🧠 Key rule

```
New session every request = suspicious
```

🐞 “Works once then fails” reason

- First request probation
- Next requests:
  - cookies missing
  - session reset → soft block

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8 Diagnosis Skills — MOST IMPORTANT

🔑 Core Idea

- **Fix later**
- **Diagnose first**

🧭 Diagnosis order (NEVER change)

1. Status code
2. Response size
3. Timing pattern
4. Headers
5. Cookies / session

🧠 Mental flow

```
Fail →
Status →
Size →
Pattern →
Headers →
Session
```

🧩 Common classifications

- 403 / 429 → Hard block
- 200 + small HTML → Soft block
- Bursts → Rate-based
- Static headers → Fingerprint
- No cookies → Session issue

🚫 Biggest mistake

- Code change before diagnosis

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9 Scraper Health Check System

🧠 Purpose

- Separate **diagnostic tool**
- Scraper ka part nahi
- Reusable for every site

🛠️ What it logs

- Status code
- Response size
- Time gap
- Header count + UA
- Cookies present or not
- Clear diagnosis message

🔁 How you use it

- Scraper fails → STOP
- Run health check on 1–3 URLs
- Read diagnosis
- Then redesign scraper

📐 STRUCTURE OF SCRAPER HEALTH CHECK SCRIPT

1️⃣ Config Section

```
URLS = [
    "problematic_url_here"
]
```

2️⃣ State

- `requests.Session()`
- `last_request_time`

3️⃣ Metrics Collected per Request

- `status_code`
- `len(response.text)`
- `time gap`
- `len(headers)`
- `cookies present`

4️⃣ Diagnosis Rules

- `403 / 429` → Hard block
- `200 + small HTML` → Soft block
- `gap < threshold` → Rate-based risk
- `few headers` → Fingerprint risk
- `no cookies` → Session issue

5️⃣ Output

- Human-readable diagnosis:

```
DIAGNOSIS:
- SOFT BLOCK
- RATE-BASED
- SESSION ISSUE
```

☑ FINAL STAGE-3 OUTCOME

Ab tum:

- panic nahi karte
- blindly fix nahi karte
- problem ka naam lagate ho
- calmly redesign karte ho

👉 **This is professional scraping mindset.**
